

Ex. No.: 8
PRODUCER CONSUMER USING SEMAPHORES

Aim:

To write a program to implement solution to producer consumer problem using semaphores.

Algorithm:

1. Initialize semaphore empty, full and mutex.
2. Create two threads- producer thread and consumer thread.
3. Wait for target thread termination.
4. Call sem_wait on empty semaphore followed by mutex semaphore before entry into critical section.
5. Produce/Consume the item in critical section.
6. Call sem_post on mutex semaphore followed by full semaphore
7. before exiting critical section. 8. Allow the other thread to enter its critical section.
9. Terminate after looping ten times in producer and consumer Threads each.

Program Code:

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>
sem_t empty, full, mutex;
int buffer[MAX];
int in = 0, out = 0;
void *producer(void *arg) {
    int item;
    for(int i = 0; i < 10; i++) {
        item = i + 1;
        sem_wait(&empty);
        sem_wait(&mutex);
        buffer[in] = item;
        printf("Produced: %d\n", item);
        in = (in + 1) % MAX;
        sem_post(&mutex);
        sem_post(&full);
        sleep(1);
    }
    return NULL;
}
void *consumer(void *arg) {
    int item;
    for(int i = 0; i < 10; i++) {
        sem_wait(&full);
```

```

sem_wait(&mutex);
item = buffer[out];
printf("Consumed: %d\n", item);
out = (out + 1) % MAX;
sem_post(&mutex);
sem_post(&empty);
sleep(1);
}
return NULL;
}
int main() {
pthread_t p, c;
sem_init(&empty, 0, MAX);
sem_init(&full, 0, 0);
sem_init(&mutex, 0, 1);
pthread_create(&p, NULL, producer, NULL);
pthread_create(&c, NULL, consumer, NULL);
pthread_join(p, NULL);
pthread_join(c, NULL);
sem_destroy(&empty);
sem_destroy(&full);
sem_destroy(&mutex);
return 0;
}

```

Output:

```

Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Consumed: 3
Produced: 4
Consumed: 4
Produced: 5
Consumed: 5
Produced: 6
Consumed: 6
Produced: 7
Consumed: 7
Produced: 8
Consumed: 8
Produced: 9
Consumed: 9

```

Produced: 10
Consumed: 10

Result:

Thus the program to implement a solution for producer consumer problem using semaphores has been successfully executed.